

Muhammad Arslan Tabish

Full Stack AI Engineer Python · FastAPI · React / TypeScript · LLM Systems

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Nationality Pakistani · Location Lahore, Pakistan · Relocation Open to relocation and remote · Notice 30 days

Profile

Full Stack AI Engineer, 2+ years professional, building LLM-powered products end to end — **Python/FastAPI** services and data models through to **React 19 / TypeScript** interfaces I design and ship myself. Focused on **AI systems, not AI demos**: structured output with schema validation and retry/repair, catalogue-grounded generation, guardrails and automated QA gates, model routing and per-call cost tracking on the **Anthropic Claude API**. Owns features from database to browser and deploys to production on Vercel, Railway and Docker.

Technical Skills

AI & LLM Engineering: Anthropic Claude API (Messages API, system prompts, token accounting), multi-stage AI pipelines and agentic workflows, prompt engineering and versioning, structured JSON output with schema validation and retry-with-nudge, grounded generation and RAG patterns, model routing, prompt caching, output guardrails, automated QA gates, eval harnesses, OpenCV

Frontend: React 18/19, TypeScript (strict), Vite, TanStack Start / Router / Query, Tailwind CSS with design tokens, Radix UI / shadcn, Framer Motion, React Three Fiber (3D), React Hook Form + Zod, React Native (Expo), Vitest + React Testing Library, vitest-axe, WCAG 2.1 AA, SEO, Figma-to-code

Backend: Python 3.12+ (async/await, asyncio, typing, OOP), FastAPI (Pydantic v2, dependency injection, background jobs, WebSockets), SQLAlchemy 2.0 async, Alembic, Django and Django REST Framework, REST/JSON, SOAP/XML, webhooks, third-party integrations (Anthropic, Stripe, Apify, Twelve Data), pytest, httpx

Databases: PostgreSQL (psycopg v3, asyncpg), MySQL, MongoDB (PyMongo, Motor), SQLite3; schema design, query optimisation, migrations, multi-tenant scoping

DevOps & Security: Docker Compose, Vercel, Railway, Cloudflare, AWS S3 (boto3), Git / GitHub / GitLab, code review, deployment pipelines, production monitoring; AES-256-GCM encryption at rest, JWT/OAuth, TOTP 2FA; Agile/Scrum, Stripe billing

Professional Experience

Intagleo Systems

Jul 2024 – Present

Software Engineer (Full Stack / Python)

Lahore, Pakistan

- Built secure, scalable **REST APIs in Python FastAPI** — authentication, **Pydantic** validation, data processing and business logic — across several production applications, plus **Django/DRF** services using the Django ORM with permissions and end-to-end workflows.
- Architected and optimised data across **PostgreSQL, MySQL** and **MongoDB**, writing efficient queries and migrations and holding data integrity across systems.
- Shipped real-time features over **WebSocket** for live chat and notifications, and built **data-bound front ends** in HTML/CSS/JavaScript for the digital-signage line.
- Automated internal engineering workflows (task routing, approvals, compliance reporting) in Python, and delivered in **Agile** teams with Git/GitHub code review, sprint planning and **weekly client meetings**.

Wisdom Software House

June 2023 – Aug 2023

Software Developer Intern

Lahore, Pakistan

- Built full-stack web applications with a **Python FastAPI** backend and optimised **MySQL** queries for performance.

Devsloop

Dec 2022 – Jan 2023

Software Developer Intern

Lahore, Pakistan

- Web development fundamentals, version control, debugging and team workflows.

Projects

Professional Projects — Intagleo Systems

Project Management Office (PMO) — Internal Automation Portal

FastAPI, PostgreSQL, GitHub API

- Built the full **FastAPI** backend for internal project tracking, developer task management and **automated workflow processing**, on a scalable API supporting multi-department access with role-based permissions and real-time status updates.
- Implemented an **automated code-review workflow** with routing and approval systems plus **GitHub integration**, and an **automated Excel reporting** system for compliance tracking and performance analytics.

Acertara — Admin Panel Backend

Django, Django REST Framework

- Built secure, scalable REST APIs with **Django** and **DRF**, using the Django ORM, CBVs and FBVs to implement complex business logic and workflow automation.

- Implemented role-based access control with **Permissions, Groups** and custom authorisation rules; built file uploads, attachment handling and **automated email integrations**; wrote **unit tests**, optimised ORM queries and ran **weekly client meetings** tracked in GitLab.

OMNI Portal — Backend Extension Services

FastAPI, MongoDB, MySQL, AWS S3

- Developed a layered **FastAPI** service (routes / controllers / services / repositories) backed by **MongoDB (Motor)** and **MySQL (aiomysql)**.
- Implemented **TOTP-based two-factor authentication** (pyotp + QR provisioning), transactional email via fastapi-mail, **AWS S3** file handling with boto3, and structured logging.

Clarity Layout Studio — Digital Signage Layout Tooling

Node.js, Express, WebSocket, HTML/CSS

- Contributed to a **layout development and live-preview workspace** for data-driven menu boards used by national brands — **Subway, Popeyes, Tim Hortons, Dairy Queen, Zaxby's** — on an **Express + WebSocket** dev server hot-reloading the board in an iframe preview.
- Built **data-bound board layouts** (data-role / data-field slot binding against slotData.xlsx) on fixed 1920×1080 / 1080×1920 canvases, joining layout HTML/CSS with Excel product data and per-client media, plus theme and scaffolding scripts.

AI Systems — self-built products

Cold Outreach Pipeline — AI Lead-Generation SaaS

Python, FastAPI, Claude, PostgreSQL, Stripe

- Designed a **9-stage, status-driven, resumable AI pipeline** (extract, enrich, AI filter, gate, email, verify, AI copy, QA, sync) where every lead advances only on a valid stage transition, so runs crash and resume without duplicating work or LLM spend.
- Built a **defensive structured-output layer** for all AI calls — fence stripping, JSON recovery, schema validation and **retry with a corrective nudge** before a typed ParseError — plus deterministic guardrails and an automated QA stage that blocks bad copy before it reaches a prospect.
- Implemented **multi-tenant isolation, BYOK API keys encrypted at rest with AES-256-GCM**, Stripe checkout with plan limits, per-call token-cost attribution, and a **zero-spend demo mode** using a mock LLM behind a Python Protocol.

Dukandar AI — Voice-First Conversational Shopping Agent

Python, FastAPI, Claude, SQLAlchemy, React

- Built a **catalogue-aware conversational agent** that walks shoppers through a facet-driven dialogue (category → kind → colour → size → price) and mirrors its filters live onto the storefront grid.
- **Constrained Claude slot extraction to facet values that exist in the live catalogue**, eliminating hallucinated and out-of-stock suggestions, with a deterministic keyword parser as a \$0 fallback on missing key, timeout or malformed JSON — the conversation never breaks.
- Built the session engine, faceted search, a **TTS proxy** for voice output and the **Libaas storefront front end** on async SQLAlchemy 2.0 + PostgreSQL.

GoldWatch — AI Gold (XAUUSD) Trading Analyst

Python, Claude, Market Data APIs

- Built a market-analysis assistant that computes market structure and indicators, renders charts, gathers macro context and asks **Claude** for a read — with one hard safety rail: **it never places an order**, it prints a full briefing (entry, stop, target, lot size, exact money at risk) and waits for the human.
- Architected the system to **resist LLM over-eagerness**: a deterministic scanner scores setups first and an API call is spent only above a configurable threshold, keeping cost near zero on quiet days.

Full-Stack & Frontend — client delivery

Khaayaal — Elder-Care Platform (Client)

TanStack Start, React 19, TypeScript, Tailwind v4

- Designed and shipped the complete six-route marketing site on **TanStack Start + Router** with **React 19** and **Tailwind CSS v4**, on a custom design-token system, **Radix UI** primitives and a reusable brand component library.
- Implemented an animated **3D hero** with **React Three Fiber**, scene-based scroll animation and an **interactive pricing calculator** over tiered care plans; self-hosted variable fonts, **React Hook Form + Zod**, deployed on **Vercel**.

Leads In The Pipe — B2B Agency Website (Client)

React 18, TypeScript, Tailwind, Framer Motion, FastAPI

- Translated a **Figma design** into a production single-page front end on **Vite + React 18 + TypeScript (strict)** with Tailwind design tokens and **Framer Motion**, backed by a **Python FastAPI** serverless API for form handling on Vercel.
- Enforced quality with **Vitest + React Testing Library + vitest-axe**, a **WCAG 2.1 AA** pass, **SEO** work (meta/OG, JSON-LD, Core Web Vitals) and a **performance budget**, on a documented PRD / TRD workflow.

Education & Certifications

University of Punjab, Lahore — BSc Computer Science (BSCS)

2020 – 2024

Certifications: **Udemy** — Python for FastAPI: Advanced Backend Development · **Udemy** — Python for Computer Vision and Deep Learning · **Battle Byte Programming Competition** — Runner-Up (university level)

Personal Details

Languages: English (professional, fluent) · Urdu (native) · Punjabi (native)

Driving Licence: Pakistan (LTV)

References available on request